

Business Statistics - MS6107E

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1 Probability Theory [1]

- Foundations of Probability
- Probability Space and Axioms
- Rules and Bounds of Probability
- Counting Techniques
- Conditional Probability and Bayesian Reasoning



“You can, for example, never foretell what any one man will do, but you can say with precision what an average number will be up to. Individuals vary, but percentages remain constant.”

— **Sherlock Holmes**

The Sign of Four, Arthur Conan Doyle



Basic Concepts of Probability

Random Experiment

An experiment that can produce **different outcomes** even when repeated under the same conditions.

Sample Space (S)

The set S of **all possible outcomes** of a random experiment.

Types of Sample Space

- **Discrete:** Finite or countably infinite outcomes.
- **Continuous:** Contains an interval of real numbers.

Event (A)

An **event** is any subset of the sample space:

$$A \subseteq S$$



Basic Concepts of Probability

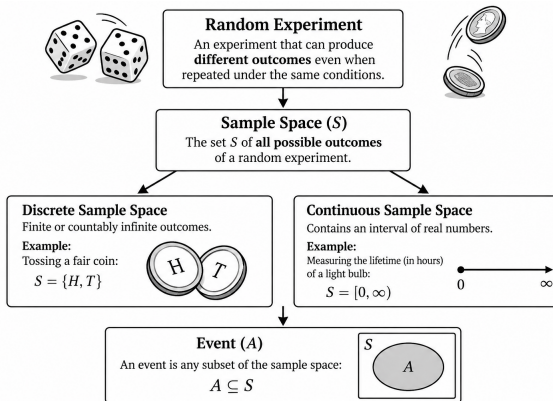


Figure: Random Experiment



Operations on Events

Union

The **union** of two events contains all outcomes that belong to E_1 , E_2 , or both.

$$E_1 \cup E_2 = \{x : x \in E_1 \text{ or } x \in E_2\}$$

Intersection

The **intersection** of two events contains all outcomes that belong to both E_1 and E_2 .

$$E_1 \cap E_2 = \{x : x \in E_1 \text{ and } x \in E_2\}$$

Complement

The **complement** of an event E contains all outcomes in the sample space S that do **not** belong to E .

$$E^c = S \setminus E = \{x \in S : x \notin E\}$$



Mutually Exclusive Events and Partition

Definition

Two events E_1 and E_2 are **mutually exclusive or disjoint** if they **cannot occur simultaneously**.

$$E_1 \cap E_2 = \emptyset$$

Definition

A collection of events

$$A_1, A_2, \dots, A_n$$

is said to form a **partition** of the sample space S if the following conditions are satisfied.

1. The events are pairwise disjoint: $A_i \cap A_j = \emptyset$ for all $i \neq j$
2. Together, the events cover the entire sample space: $\bigcup_{i=1}^n A_i = S$



Commutative Laws

Commutative Law

The **order of events does not matter** when taking their union or intersection.

For union,

$$E_1 \cup E_2 = E_2 \cup E_1$$

For intersection,

$$E_1 \cap E_2 = E_2 \cap E_1$$



Associative Laws

Associative Law

When three or more events are involved, the **grouping of events does not matter**.

For union,

$$(E_1 \cup E_2) \cup E_3 = E_1 \cup (E_2 \cup E_3)$$

For intersection,

$$(E_1 \cap E_2) \cap E_3 = E_1 \cap (E_2 \cap E_3)$$



Distributive Laws

Distributive Law

Union and intersection can be **distributed over each other**.

Intersection distributes over union:

$$E_1 \cap (E_2 \cup E_3) = (E_1 \cap E_2) \cup (E_1 \cap E_3)$$

Union distributes over intersection:

$$E_1 \cup (E_2 \cap E_3) = (E_1 \cup E_2) \cap (E_1 \cup E_3)$$



De Morgan's Laws

First De Morgan's Law

The complement of a union is the intersection of the complements.

$$(E_1 \cup E_2)^c = E_1^c \cap E_2^c$$

Second De Morgan's Law

The complement of an intersection is the union of the complements.

$$(E_1 \cap E_2)^c = E_1^c \cup E_2^c$$



Sigma Algebra (σ -Algebra)

Definition

Let S be a sample space. A collection $\mathcal{F}$ of subsets of S is called a **σ -algebra** if it satisfies the following properties.

- 1 The sample space belongs to $\mathcal{F}$:

$$S \in \mathcal{F}$$

- 2 If an event belongs to $\mathcal{F}$, its complement also belongs to $\mathcal{F}$:

$$A \in \mathcal{F} \implies A^c \in \mathcal{F}$$

- 3 $\mathcal{F}$ is closed under countable unions:

$$A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$$

$\mathcal{F}$ is the collection of **events to which probabilities can be assigned**.



Definition

Given a sample space S and a σ -algebra $\mathcal{F}$, a **probability measure** is a function $P : \mathcal{F} \rightarrow [0, 1]$ that assigns a probability $P(A)$ to every event $A \in \mathcal{F}$. It satisfies the following axioms:

1 Non-negativity

$$P(A) \geq 0$$

2 Normalization

$$P(S) = 1$$

3 Countable Additivity

For pairwise disjoint events $A_1, A_2, \dots$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

$$(S, \mathcal{F}, P) = \text{Probability Space}$$



σ Algebra - 2 Coins

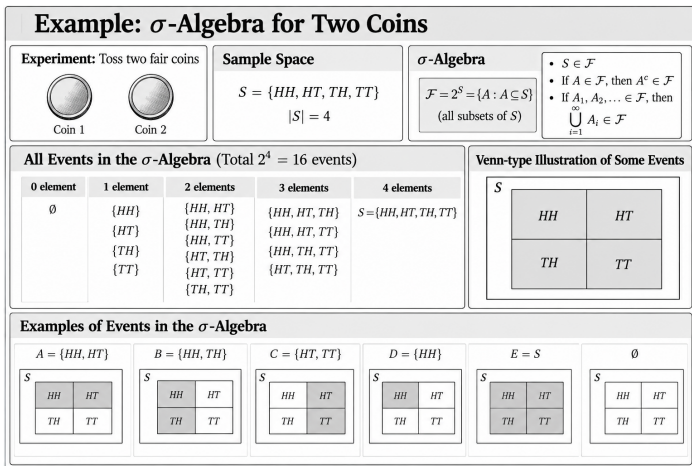


Figure: σ Algebra - 2 Coins



Finite Additivity of Probability

Finite Additivity

Let $A_1, A_2, \dots, A_n$ be **pairwise disjoint** events, i.e.,

$$A_i \cap A_j = \emptyset \quad \text{for } i \neq j.$$

Then,

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

For two mutually exclusive events A and B ,

$$A \cap B = \emptyset$$

and hence,

$$P(A \cup B) = P(A) + P(B)$$



Probability of the Empty Set

From the normalization axiom,

$$P(S) = 1.$$

Since

$$S \cup \emptyset = S$$

and S and $\emptyset$ are disjoint, finite additivity gives

$$P(S \cup \emptyset) = P(S) + P(\emptyset).$$

Therefore,

$$P(S) = P(S) + P(\emptyset).$$

Hence,

$$P(\emptyset) = 0$$



Probability of a Complement

Let A be an event in the sample space S .

Since

$$A \cap A^c = \emptyset$$

and

$$A \cup A^c = S$$

finite additivity gives

$$P(A \cup A^c) = P(A) + P(A^c)$$

But

$$P(S) = 1$$

Therefore,

$$1 = P(A) + P(A^c)$$



Probability of $B \cap A^c$

The event B can be partitioned into two disjoint parts:

$$B = (A \cap B) \cup (B \cap A^c).$$

Also,

$$(A \cap B) \cap (B \cap A^c) = \emptyset$$

Therefore, by finite additivity,

$$P(B) = P(A \cap B) + P(B \cap A^c)$$

Rearranging,

$$P(B \cap A^c) = P(B) - P(A \cap B)$$



Addition Rule of Probability

Consider two arbitrary events A and B .

We can write

$$A \cup B = A \cup (B \cap A^c)$$

The events A and $B \cap A^c$ are disjoint. Therefore,

$$P(A \cup B) = P(A) + P(B \cap A^c)$$

Using

$$P(B \cap A^c) = P(B) - P(A \cap B)$$

we obtain

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



Addition Rule: Special Case

For any two events,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If A and B are mutually exclusive,

$$A \cap B = \emptyset.$$

Therefore,

$$P(A \cap B) = 0.$$

Hence,

$$P(A \cup B) = P(A) + P(B)$$

Key Point

Finite additivity is therefore a special case of the general addition rule when the events are mutually exclusive.



Monotonicity of Probability

Result

If

$$A \subseteq B,$$

then

$$P(A) \leq P(B)$$

Since $A \subseteq B$, we can write

$$B = A \cup (B \cap A^c)$$

These two events are disjoint.

Therefore,

$$P(B) = P(A) + P(B \cap A^c)$$

Since probability $P(B \cap A^c)$ is non-negative,

$$P(B) \geq P(A)$$



Boole's Inequality

From the addition rule,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Since

$$P(A \cap B) \geq 0$$

we obtain

$$P(A \cup B) \leq P(A) + P(B)$$

More generally,

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i)$$



Bonferroni Inequality

For two events A and B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Since

$$P(A \cup B) \leq 1$$

we have

$$P(A) + P(B) - P(A \cap B) \leq 1$$

Rearranging,

$$P(A \cap B) \geq P(A) + P(B) - 1$$



Bonferroni Inequalities for Two Events

For two events A and B , useful bounds are:

Upper bound for the union:

$$P(A \cup B) \leq P(A) + P(B)$$

Lower bound for the intersection:

$$P(A \cap B) \geq P(A) + P(B) - 1$$

Since probability cannot be negative,

$$P(A \cap B) \geq \max\{0, P(A) + P(B) - 1\}$$

Also,

$$P(A \cap B) \leq \min\{P(A), P(B)\}$$



Bounds for Union and Intersection

For any two events A and B ,

Intersection

$$\max\{0, P(A) + P(B) - 1\} \leq P(A \cap B) \leq \min\{P(A), P(B)\}$$

Union

$$\max\{P(A), P(B)\} \leq P(A \cup B) \leq \min\{1, P(A) + P(B)\}$$

These bounds are useful when $P(A)$ and $P(B)$ are known but $P(A \cap B)$ or $P(A \cup B)$ is not known.



Counting Techniques

Why Counting?

Probability often requires us to determine the number of **possible outcomes** and the number of **favourable outcomes**.

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

Technique	When to Use
Multiplication Principle	Sequence of choices
Permutation	Order matters
Combination	Order does not matter
Permutation with Repetition	Some objects are identical
Matrix / Table Method	Two or more categories of choices



Multiplication Principle

Multiplication Principle and Sequence of Choices

Multiplication Principle

Suppose a process consists of a **sequence of choices**.

If

- the first choice can be made in n_1 ways,
- the second choice can be made in n_2 ways,
- ...
- the k th choice can be made in n_k ways,

then the total number of possible outcomes is

$$N = n_1 n_2 \cdots n_k = \prod_{i=1}^k n_i$$



Permutation

Order Matters

A **permutation** is an arrangement of objects where **order matters**. The number of ways of arranging r objects selected from n distinct objects is

$${}^n P_r = \frac{n!}{(n-r)!}$$

If all n objects are arranged,

$${}^n P_n = n!$$

Example: From 5 employees, how many ways can we select a **President, Secretary, and Treasurer**?

$${}^5 P_3 = \frac{5!}{(5-3)!} = 5 \times 4 \times 3 = \boxed{60}$$

The positions are different, so **order matters**.



Combination

Order Does Not Matter

A **combination** is a selection of objects where **order does not matter**. The number of ways of selecting r objects from n distinct objects is

$${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Example: From 5 employees, how many different **3-member committees** can be formed?

$${}^5C_3 = \frac{5!}{3!2!} = \boxed{10}$$

Here,

$$\{A, B, C\} = \{C, A, B\}$$

so **order does not matter**.



Permutation of Similar Objects

When Some Objects are Identical

Suppose n objects contain identical objects: $n_1, n_2, \dots, n_k$ where $n_1 + n_2 + \dots + n_k = n$

The number of distinct arrangements is

$$\frac{n!}{n_1! n_2! \dots n_k!}$$

Example: LEVEL

The word **LEVEL** contains 5 letters:

$L, L, \quad E, E, \quad V$

$$\text{Number of arrangements} = \frac{5!}{2!2!} = \boxed{30}$$



Counting Arrangements

Selecting r Objects from n Objects

Suppose we have n distinct objects and want to select or arrange r objects.

Two questions determine the appropriate counting rule:

Question 1: Does Order Matter?

Is

$$(A, B) \neq (B, A)?$$

If yes or order matters, we are interested in an **arrangement**.

If no or order matters, we are interested only in a **selection**.

Question 2: Is Repetition Allowed?

Can the same object be selected more than once? If yes, it allows **replacement**.

If no, it does not allow **replacement**.



Case I: Order Matters, Repetition Allowed

Suppose we select r objects from n distinct objects.

Conditions

- Order **matters**.
- Repetition **is allowed**.

For every position, there are n possible choices:

$$\underbrace{n}_1 \times \underbrace{n}_2 \times \underbrace{n}_3 \times \cdots \times \underbrace{n}_r$$

Therefore,

$$N = n^r$$

Example: A 4-digit PIN is constructed using the digits 0–9, with repetition allowed - $10^4 = 10,000$ possible PINs.



Case II: Order Matters, No Repetition

Suppose we select and arrange r objects from n distinct objects without replacement.

Conditions

- Order **matters**.
- Repetition is **not allowed**.

The number of choices decreases at each stage:

$$n \times (n - 1) \times (n - 2) \times \cdots \times (n - r + 1).$$

Therefore,

$${}_nP_r = \frac{n!}{(n - r)!}$$

Example: From 10 employees, choose a President, Vice-President and Secretary.

$${}_{10}P_3 = \frac{10!}{7!} = 10 \times 9 \times 8 = 720$$



Why Does Order Matter in Permutations?

Suppose we select two people from

$$\{A, B, C\}$$

for the positions of President and Secretary.

The possible arrangements are

$$(A, B), (A, C), (B, A), (B, C), (C, A), (C, B).$$

Therefore,

$${}_3P_2 = 3 \times 2 = 6.$$

Notice that

$$(A, B) \neq (B, A)$$



Case III: Order Does Not Matter, No Repetition

Suppose we select r objects from n distinct objects.

Conditions

- Order does **not** matter.
- Repetition is **not allowed**.

If order mattered, there would be nP_r arrangements.
But each group of r objects is counted

$$r!$$

times because its members can be rearranged in $r!$ ways.
Therefore,

$$\boxed{\binom{n}{r} = \frac{nP_r}{r!} = \frac{n!}{r!(n-r)!}}$$



Combination: Example

Suppose a committee of 3 people must be selected from 10 employees. The order in which the members are selected does not matter. For example,

$$\{A, B, C\} = \{C, A, B\}.$$

Therefore,

$$\begin{aligned}\binom{10}{3} &= \frac{10!}{3!7!} \\ &= \frac{10 \times 9 \times 8}{3 \times 2 \times 1}\end{aligned}$$

$$\boxed{\binom{10}{3} = 120}$$



Case IV: Order Does Not Matter, Repetition Allowed

Suppose we select r objects from n types of objects.

Conditions

- Order does **not** matter.
- Repetition **is allowed**.

This is called a **combination with repetition**.

The number of possible selections is

$$\binom{n + r - 1}{r}$$

or equivalently,

$$\binom{n + r - 1}{n - 1}$$



Combination with Repetition

Example

A shop sells three types of doughnuts:

Chocolate, Vanilla, Strawberry.

Suppose a customer buys 5 doughnuts.

The same type can be selected repeatedly, and the order of selection does not matter.

Thus,

$$n = 3, \quad r = 5.$$

The number of possible selections is

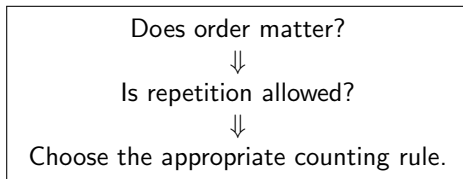
$$\begin{aligned}\binom{n+r-1}{r} &= \binom{3+5-1}{5} \\ &= \binom{7}{5} = \binom{7}{2}\end{aligned}$$



Counting Arrangements

Summary Matrix

	Repetition Allowed	No Repetition
Order Matters	n^r	${}_nP_r = \frac{n!}{(n-r)!}$
Order Does Not Matter	$\binom{n+r-1}{r}$	$\binom{n}{r} = \frac{n!}{r!(n-r)!}$



Conditional Probability

Idea

Conditional probability asks: **How does the probability of B change when we know that A has occurred?**

It is denoted by

$$P(B | A)$$

and defined as

$$P(B | A) = \frac{P(A \cap B)}{P(A)} \quad P(A) > 0$$

Knowing that A occurred effectively changes our sample space from S to A .

Example: A = Customer purchased a laptop, B = Customer purchased an extended warranty. $P(B | A) = P(\text{Warranty} | \text{Laptop})$ is the probability of purchasing a warranty given that the customer purchased a laptop.



Multiplication Rule

Starting with conditional probability,

$$P(B | A) = \frac{P(A \cap B)}{P(A)}$$

Rearranging,

$$P(A \cap B) = P(A)P(B | A)$$

Similarly,

$$P(A \cap B) = P(B)P(A | B)$$

Therefore,

$$P(A)P(B | A) = P(B)P(A | B)$$



Law of Total Probability

Suppose A and A^c divide the sample space.

Then event B can occur in two mutually exclusive ways:

$$B = (A \cap B) \cup (A^c \cap B)$$

Therefore,

$$P(B) = P(A \cap B) + P(A^c \cap B)$$

Using the multiplication rule,

$$P(B) = P(B | A)P(A) + P(B | A^c)P(A^c)$$

More generally, if $A_1, A_2, \dots, A_k$
form a partition of the sample space,

$$P(B) = \sum_{i=1}^k P(B | A_i)P(A_i)$$



Concept of Independence

Two events A and B are **independent** if knowing that one event occurred does not change the probability of the other.

Thus,

$$P(B | A) = P(B)$$

or equivalently,

$$P(A | B) = P(A)$$

Using the multiplication rule,

$$P(A \cap B) = P(A)P(B | A)$$

and therefore, for independent events,

$$P(A \cap B) = P(A)P(B)$$

Mutually exclusive events cannot occur together; independent events do not influence each other's probabilities.



Bayes' Theorem

From conditional probability,

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Using

$$P(A \cap B) = P(B | A)P(A),$$

we obtain

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

Using the Total Probability Rule,

$$P(A | B) = \frac{P(B | A)P(A)}{P(B | A)P(A) + P(B | A^c)P(A^c)}$$



Understanding Bayes' Theorem

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

Term	Meaning
$P(A)$	Prior probability
$P(B A)$	Likelihood
$P(B)$	Marginal probability / Evidence
$P(A B)$	Posterior probability

Conceptually,

Prior Belief + New Evidence $\longrightarrow$ Updated Belief

Bayesian Thinking

Bayes' theorem provides a mathematical mechanism for **updating probabilities when new information becomes available**.



References I



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Statistical inference.
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